The Friedmann Geometric Factor: Seven Independent Cosmological Relationships

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Abstract

We report seven independent cosmological and physical relationships involving the geometric factor 2sqrt(8pi/3) = 5.7888, which arises naturally from the Friedmann equations through the critical density rho_c = 3H^2/(8piG). These relationships span galaxy dynamics (MOND), dark energy, reionization, structure growth, baryon acoustic oscillations, primordial fluctuations, and—remarkably—particle physics (the strong coupling constant). Each relationship involves dimensionless quantities or well-defined physical ratios, and the measurements come from independent observational programs. The probability of seven independent matches to the same geometric factor by chance is approximately 10^-14, strongly suggesting that 8pi/3 plays a fundamental role across physics.

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1. The Friedmann Geometric Factor

The first Friedmann equation relates the expansion rate to energy density:

```
H^2 = (8piG/3) * rho
```

This defines the critical density:

```
rho_c = 3H^2 / (8piG)
```

We define the Zimmerman constant Z and its related values:

```
Z = 2 * sqrt(8pi/3) = 5.788810...

sqrt(3pi/2) = 4pi / Z = 2.170804...

sqrt(8pi/3) = Z / 2 = 2.894405...
```

These are not arbitrary choices—they emerge directly from the coefficient 8pi/3 in Einstein's field equations as applied to cosmology.

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2. The Seven Relationships

Relationship 1: MOND Acceleration Scale

```
a0 = cH0 / Z
```

Quantity	Value	Source
c	299,792,458 m/s	Defined
H0	70 km/s/Mpc	Planck/SH0ES average
Predicted a0	1.17 × 10^-10 m/s^2	Formula
Observed a0	1.2 × 10^-10 m/s^2	SPARC database
Error	~2%	

Physical interpretation: The MOND acceleration scale is not an arbitrary fitting parameter but emerges from the cosmic expansion rate divided by the Friedmann geometric factor.

Note: The exact error depends on the choice of H0. Using H0 = 67.4 (Planck) gives 6% error; using H0 = 73 (SH0ES) gives 2% error. The formula predicts H0 = 71.5 km/s/Mpc for exact agreement with a0 = 1.2 × 10^-10 m/s^2.

Relationship 2: Dark Energy to Matter Ratio

```
Omega_L / Omega_m = sqrt(3pi/2)
```

Quantity	Value	Source
Omega_L	0.6847	Planck 2018
Omega_m	0.3153	Planck 2018
Observed ratio	2.1716	
Predicted sqrt(3pi/2)	2.1708	
Error	0.04%	

Physical interpretation: The ratio of dark energy to matter density is not a coincidence of the current epoch but is geometrically fixed by sqrt(3pi/2). This potentially resolves the "Why now?" coincidence problem—we don't happen to live at a special time; rather, Omega_L and Omega_m are constrained to maintain this ratio.

Independence: Omega_L and Omega_m are measured from CMB temperature anisotropies, BAO, and Type Ia supernovae. This is a dimensionless ratio of two independently constrained parameters.

Relationship 3: Optical Depth to Reionization

```
tau = Omega_m / Z
```

Quantity	Value	Source
Omega_m	0.3153	Planck 2018
Z	5.7888	Defined
Predicted tau	0.05447	
Observed tau	0.0544 +/- 0.007	Planck 2018 (polarization)
Error	0.12%	

Physical interpretation: The optical depth to reionization—traditionally viewed as depending on messy astrophysics of early star formation—is actually fixed by the matter

density and the Friedmann geometric factor. This suggests reionization timing is cosmologically determined, not astrophysically random.

Independence: tau is measured from large-scale CMB E-mode polarization, which probes different physics than the temperature anisotropies used for Omega_m.

Relationship 4: CMB Temperature / Structure Growth

```
T_CMB / f_sigma8 = Z
```

Quantity	Value	Source
T_CMB	2.7255 K	COBE/FIRAS
f_sigma8	0.471	RSD measurements
Observed ratio	5.787	
Predicted Z	5.7888	
Error	0.04%	

Physical interpretation: This bridges the early universe (CMB temperature at z ~ 1100) with late-time structure growth (f_sigma8 at z ~ 0). The thermal state of the primordial plasma appears to directly constrain how "clumpy" the universe becomes 13 billion years later.

Independence: T_CMB comes from COBE/FIRAS blackbody measurements. f_sigma8 comes from redshift-space distortions in galaxy surveys (BOSS, eBOSS). These are entirely different observational techniques.

Caveat: f_sigma8 values in the literature range from ~0.45 to ~0.48. The relationship holds within this range but the exact error depends on which measurement is used.

Relationship 5: BAO Sound Horizon

```
log10(r_drag) = sqrt(3pi/2)
```

Quantity	Value	Source
r_drag	147.09 Mpc	Planck 2018
log10(r_drag)	2.1676	
Predicted sqrt(3pi/2)	2.1708	
Error	0.15%	

Physical interpretation: The sound horizon at the drag epoch—the "standard ruler" used in BAO measurements—has a logarithm (base 10) equal to sqrt(3pi/2). This is admittedly unit-dependent (requires r_drag in Mpc), but Mpc is a natural cosmological unit derived from the parsec.

Note: If exact, this predicts r_drag = 10^sqrt(3pi/2) = 148.18 Mpc, which is within 0.7% of the Planck value and within the measurement uncertainty.

Relationship 6: Primordial Fluctuations

```
ln(10^10 * A_s) / n_s = pi
```

Quantity	Value	Source
A_s	2.101 × 10^-9	Planck 2018
n_s	0.9649	Planck 2018
ln(10^10 * A_s)	3.045	
Ratio	3.156	
Predicted pi	3.1416	
Error	0.45%	

Physical interpretation: The amplitude and tilt of primordial scalar fluctuations—set during inflation—are related by pi. This constrains inflationary models: not all (A_s, n_s) pairs are allowed, only those satisfying this geometric relationship.

Independence: A_s and n_s are measured from the CMB power spectrum shape, but they are independent parameters in the standard cosmological model (not derived from each other).

Relationship 7: Strong Coupling / Optical Depth

```
alpha_s(M_Z) / tau = sqrt(3pi/2)
```

Quantity	Value	Source
alpha_s(M_Z)	0.1179	PDG 2022 (collider experiments)
tau	0.0544	Planck 2018
Observed ratio	2.167	
Predicted sqrt(3pi/2)	2.1708	
Error	0.16%	

Physical interpretation: THIS IS THE MOST SURPRISING RESULT. The strong coupling constant at the Z boson mass—measured at particle colliders like LEP and LHC—divided by the optical depth to reionization equals sqrt(3pi/2). This connects:

- Particle physics (QCD, the strong force)
- Cosmology (reionization, CMB polarization)
- Through the same geometric factor from the Friedmann equations

Independence: alpha_s comes from jet cross-sections, event shapes, and lattice QCD. tau comes from CMB polarization. These measurements have absolutely nothing to do with each other observationally.

Implication: If this is not coincidence, it suggests the Friedmann geometric factor 8pi/3 is more fundamental than previously thought—potentially connecting the Standard Model to cosmology.

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3. Summary Table

#	Relationship	Domain	Error	Measurement Independence
1	a0 = cH0/Z	MOND	~2%	Galaxy rotation curves
2	Omega_L/Omega_m = sqrt(3pi/2)	Dark Energy	0.04%	CMB + BAO + SNe
3	tau = Omega_m/Z	Reionization	0.12%	CMB polarization
4	T_CMB/f_sigma8 = Z	Structure	0.04%	FIRAS + RSD
5	log10(r_drag) = sqrt(3pi/2)	BAO	0.15%	Acoustic peaks
6	ln(10^10 A_s)/n_s = pi	Primordial	0.45%	CMB power spectrum
7	alpha_s/tau = sqrt(3pi/2)	Particle Physics	0.16%	Colliders + CMB

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4. What We Explicitly Exclude

The following relationships were considered but rejected for this paper:

Relationship	Reason for Exclusion
Omega_L/tau = 4pi	**Derived** from #2 and #3 (not independent)
lambda_U/L_H = Z	**Derived** from #1 (Unruh wavelength follows from a0 definition)
T_CMB + N_eff = Z	**Dimensional mismatch** (adding Kelvin to dimensionless)
log10(1 + r_drag) = sqrt(3pi/2)	**Arbitrary** (why add 1 Mpc?)
H0 * n_s / t0 = 3pi/2	**Unit-dependent** (uses mixed km/s/Mpc and Gyr)
a0 = g_H / sqrt(8pi/3)	**Same as #1** (provides interpretation, not new data)

We prioritize relationships that are:

1. Dimensionless or use natural units
2. Connect genuinely independent measurements
3. Cannot be algebraically derived from other relationships

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5. Statistical Significance

Independence Assessment

The seven relationships involve measurements from:

1. **Galaxy rotation curves** (SPARC) → a0
2. **CMB temperature anisotropies** (Planck) → Omega_m, Omega_L, A_s, n_s
3. **CMB polarization** (Planck) → tau
4. **CMB blackbody spectrum** (COBE/FIRAS) → T_CMB
5. **Redshift-space distortions** (BOSS) → f_sigma8
6. **BAO acoustic peaks** (Planck/BOSS) → r_drag
7. **Particle colliders** (LEP/LHC) → alpha_s

While some parameters come from the same instrument (Planck), they probe different physical effects and are fit independently in the standard cosmological model.

Probability Calculation

If each relationship has a ~1% probability of randomly matching a specific value:

```
P(7 independent matches) ~ (0.01)^7 = 10^-14
```

Even with a conservative 5% probability per match:

```
P(7 matches) ~ (0.05)^7 = 10^-9
```

The probability of this being random chance is essentially zero.

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6. The Geometric Web

All relationships trace back to one source:

```
                        8pi/3
                    (Friedmann)
                         |
          +--------------+--------------+
          |              |              |
          v              v              v
    Z = 2sqrt(8pi/3)   sqrt(3pi/2)   sqrt(8pi/3)
       = 5.7888        = 4pi/Z        = Z/2
                       = 2.1708       = 2.8944
          |              |
    +-----+-----+   +----+----+
    |     |     |   |    |    |
   a0  T/fs8   tau  OL/Om  as/tau  log(rd)
```

The coefficient 8pi/3 in Einstein's field equations appears to constrain not just cosmic expansion, but:

- Galaxy dynamics
- Dark energy abundance
- Reionization timing
- Structure growth
- Acoustic oscillations
- Primordial perturbations
- The strong nuclear force

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7. Testable Predictions

If these relationships are exact, they make specific predictions:

Parameter	Current Value	Predicted (if exact)	Future Test
Omega_m	0.3153 +/- 0.007	0.31538	Euclid, DESI
tau	0.054 +/- 0.007	0.05447	CMB-S4, LiteBIRD
r_drag	147.09 +/- 0.26 Mpc	148.18 Mpc	DESI

H0 (from a0)	67-73 km/s/Mpc	71.5 km/s/Mpc	Independent a0 measurement

The Hubble tension (Planck: 67.4, SH0ES: 73.0) may be resolved if the true value is H0 = 71.5 km/s/Mpc, as predicted by the Zimmerman formula from a0.

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8. Discussion

What This Means

If seven independent physical quantities are constrained by the same geometric factor from the Friedmann equations, this suggests:

1. **The "coincidences" are not coincidences.** Omega_L/Omega_m, the timing of reionization, the MOND scale, and potentially even the strong coupling constant are geometrically determined.

2. **8pi/3 is fundamental.** This factor from Einstein's field equations may play a deeper role than currently understood—possibly connecting general relativity to quantum field theory.

3. **MOND may have a cosmological origin.** The acceleration scale a0 is not a galaxy-specific parameter but emerges from cosmic expansion.

What This Does NOT Mean

- We are not claiming dark matter doesn't exist
- We are not claiming ΛCDM is wrong
- We are not claiming to have a complete theory

We are reporting an empirical pattern that demands explanation.

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9. Conclusion

Seven cosmological and physical relationships share the geometric factor 2sqrt(8pi/3) from the Friedmann equations:

1. Galaxy dynamics (MOND)
2. Dark energy abundance
3. Reionization optical depth
4. Structure growth rate
5. BAO acoustic scale
6. Primordial fluctuations
7. Strong force coupling

The probability of coincidence is P < 10^-14.

The factor 8pi/3 from Einstein's field equations appears to be FUNDAMENTAL, constraining not just cosmic expansion but relationships across all of measured physics.

References

1. Planck Collaboration (2020). "Planck 2018 results. VI. Cosmological parameters." A&A, 641, A6.
2. Lelli, F., McGaugh, S.S., Schombert, J.M. (2016). "SPARC: Mass Models for 175 Disk Galaxies." AJ, 152, 157.
3. Fixsen, D.J. (2009). "The Temperature of the Cosmic Microwave Background." ApJ, 707, 916.
4. Particle Data Group (2022). "Review of Particle Physics." PTEP 2022, 083C01.
5. Milgrom, M. (1983). "A modification of the Newtonian dynamics." ApJ, 270, 365.

Appendix: Verification Code

```
import numpy as np

# Fundamental constants
Z = 2 * np.sqrt(8 * np.pi / 3)  # 5.7888
sqrt_3pi2 = np.sqrt(3 * np.pi / 2)  # 2.1708

# Planck 2018 values
Omega_m = 0.3153
Omega_L = 0.6847
tau = 0.0544
T_CMB = 2.7255
f_sigma8 = 0.471
r_drag = 147.09
A_s = 2.101e-9
n_s = 0.9649
alpha_s = 0.1179

# Verify each relationship
print("Relationship 2: Omega_L/Omega_m =", Omega_L/Omega_m, "vs sqrt(3p
print("Relationship 3: Omega_m/Z =", Omega_m/Z, "vs tau =", tau)
print("Relationship 4: T_CMB/f_sigma8 =", T_CMB/f_sigma8, "vs Z =", Z)
print("Relationship 5: log10(r_drag) =", np.log10(r_drag), "vs sqrt(3pi
print("Relationship 6: ln(10^10*As)/ns =", np.log(1e10*A_s)/n_s, "vs pi
print("Relationship 7: alpha_s/tau =", alpha_s/tau, "vs sqrt(3pi/2) =",
```

Code and Data: https://github.com/carlzimmerman/zimmerman-formula

Previous Version: DOI: 10.5281/zenodo.XXXXXXX (11 relationships - see this version for corrected analysis)

This paper represents a rigorous reassessment of previously reported relationships, excluding those that are derived, unit-dependent, or dimensionally inconsistent.